The Diverse Fates of Massive Stars in the Pair-Instability Regime: Impact of Binary Interactions

Very massive stars ($\gtrsim 100 M_{\odot}$) encounter the pair-instability limit during their evolution, which triggers their collapse and thermonuclear explosion(s) before they run out of nuclear fuel. Such stars are the progenitors of theorized (pulsational) pair-instability supernovae (P)PISN that lack unambiguous observational confirmation, and their remnants (if any) directly impact the black hole mass function actively being probed through gravitational wave observations. However, all current (P)PISN models are created with single stars, even though most massive stars have a companion with which they interact, dramatically altering the evolutionary pathway and final fate of both stars in the binary. We propose to develop the first suite of detailed, publicly available models of massive interacting binaries, where the stars encounter the pair-instability limit during their evolution. For the first time, we will robustly demonstrate the influence of binary interactions on the evolution and fate of (P)PISN progenitors. We will use these novel models to address diverse science cases, including a rigorous determination of the remnant mass distribution of (pulsational) pair-instability supernovae, and the impact of binary interactions on the signatures of associated electromagnetic transients. Our proposed program is thus timely and critical for several ongoing efforts such as: 1) accurately measuring the upper mass-gap in the black hole mass distribution through gravitational wave observations; 2) identifying transients associated with (P)PISN through various time-domain surveys like LSST; and 3) used as initial conditions for 3D simulations of (P)PISN.

Funding Call : NASA FINESST

■ Scientific Justification

The Need to Account for Binary Interactions in Models for (P)PISN Progenitors

Very massive stars ($\gtrsim 100\,M_\odot$) encounter a disruptive instability in the late stages of their evolution. Before running out of viable nuclear fuel, the low-density cores of these radiation-pressure dominated stars are hot enough to trigger the so called pair-instability, where radiation energy density is converted to electron-positron pairs (Barkat et al., 1967). The loss of radiation pressure robs the core of its pressure support, triggering its collapse, resulting in a thermonuclear explosion that leads to diverse fates depending on the mass of the star. Such stars end their lives in the long theorized (pulsational) pair-instability supernovae (Woosley, 2017). In the era of large sky surveys like LSST, it is an active effort to search for an unambiguous confirmation of the associated transients, and the remnants (if any) that they leave behind directly impact the black hole mass function that is only now being probed with gravitational wave observations (e.g., Tong et al., 2025).

However, there is a problem with all existing models of the progenitors of (P)PISN, which only model the evolution of single stars (e.g., Marchant et al., 2019). This is despite the fact that most massive stars are found to be in binaries. The binary fraction increases with mass (Offner et al., 2023), and is independent of metallicity for massive stars (Sana et al., 2025). A large fraction of binaries are in orbits close enough that the two stars will interact with each other (Sana et al., 2012). These interactions significantly alter the evolutionary pathway of both stars (Laplace et al., 2021; Renzo and Götberg, 2021), and can even result in mergers. These effects, which *will* occur in nature, have not yet been accounted for in existing (P)PISN progenitor models. To enable a complete and accurate understanding of current and future observational datasets, it is essential to incorporate the realistic effects of binarity in our theoretical models.

We propose to create the much needed detailed stellar evolution models of massive interacting binaries in the pair-instability regime. Our models will help interpret, and advance our understanding of observational datasets from several NASA missions. Using the state-of-the-art open source stellar evolution code MESA (Modules for Experiments in Stellar Astrophysics) (Jermyn et al., 2023), we will model the evolution of both stars through their interaction period, and also the pair-instability phase. *These binary evolution models will be the first of their kind*, and will allow us to robustly quantify the impacts of binary interactions on the evolution of stars that encounter the pair-instability limit, and subsequently determine the resulting remnant mass distribution. Our publicly available models will be released to the community, providing necessary tools to interpret the black hole mass function inferred through gravitational wave observations, predict signatures of transients associated with (P)PISN, and to be used as initial conditions for 3D simulations of (P)PISN explosions. This program is thus both critical and timely.

Binary Interactions Will Impact the Evolution of (P)PISN Progenitors

Most existing models of (P)PISN progenitors simulate the evolution of single, bare helium cores, assuming that any remaining envelope is lost during the first pulse, or through binary interactions (e.g., Woosley, 2017; Farag et al., 2022). However, binary interactions are expected to dominate the evolution of massive stars, including the mass-range relevant for (P)PISN progenitors (Offner et al., 2023). Binary interactions can not only remove the stellar envelope of the mass-losing donor star, but also significantly alter the internal structure of both stars in the binary. **Binary stars are not two single stars!** When compared to single stars, the stripped donors have systematically different density structures and composition profiles (Laplace et al., 2021), which are essential components to determine the outcome of pair-instability evolution. These differing structures also impact their "explodability", and lead to divergent outcomes at core-collapse (Vartanyan et al., 2021). The mass-gaining accretor star, which can also enter the pair-instability regime is not left unaffected. Its core can grow due to gaining mass, and also modify its core-envelope boundary (e.g., Renzo et al., 2023; Wagg et al., 2024). The structure, composition, and mass of a star are key determinants of the outcome of the pair-instability phase (Renzo and Smith, 2026). Existing single star models do not account for the differences caused by binarity, but they must be included for a realistic and comprehensive understanding of pair-instability evolution.

We require (P)PISN progenitor models that incorporate the full range of applicable physics, especially the impact of binary interactions that the stars will likely experience. The effects of these interactions on pair-instability evolution cannot be captured without computing detailed binary models of (P)PISN progenitors, such as those proposed here.

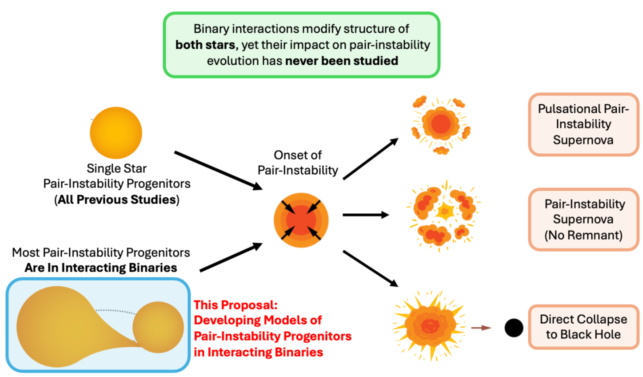

Figure 1: **New Detailed Binary Evolution Models for Progenitors of (Pulsational) Pair Instability Supernovae**. The schematic shows the different outcomes of stars that go through pair-instability evolution: 1) Pulsational Pair-Instability Supernova; 2) Pair-Instability Supernova; 3) Direct collapse to a Black Hole. However, as highlighted, all previous studies have focused on modeling the evolution of single star pair-instability progenitors, despite overwhelming observational evidence that most massive stars are in interacting binaries that can significantly alter the structure and evolution of *both stars*. This program will develop the first detailed models of pair-instability progenitors in interacting binaries, allowing us to quantify the impact of binary interactions that are important for interpreting both gravitational wave observations and electromagnetic transients associated to (pulsational) pair-instability supernovae. (Illustration Credits: partially adapted from Farag et al., 2022, Tom Wagg)

This Proposal: Developing Models for Binary Progenitors of (Pulsational) Pair-Instability Supernovae

(P)PISN are among the last theorized transients lacking an uncontroversial detection. Understanding the evolution of their progenitor stars, and their final fates is the target of several NASA observatories. To fully realize the potential of these programs, we require realistic models, which is the focus of this proposal.

We propose to develop detailed models for progenitors of (P)PISN that include the effects of binary interactions, which, although expected, have not yet been accounted for (see Fig. 1). We will utilize the open-source stellar evolution code MESA to compute self-consistent models of binary stars in the mass regime relevant for pair-instability evolution, and evolve both stars through their interaction as well as pair-instability phases. This will allow us to not only quantify the impacts of binary interactions on the structure and composition of both stars, but also evaluate their end states after pair-instability evolution. We will determine the nature of the remnants that they leave behind, and also characterize the properties of the transients associated to the explosions, both of which are directly relevant to several observables across the electromagnetic and gravitational wave spectrum.

Our models will be first to include the effects of binarity in the evolution and fate of (P)PISN progenitors. This establishes an essential step towards a comprehensive understanding of theoretical and observational aspects of (P)PISN, as, although binarity is ubiquitous across massive stars, its true impact on pair-instability has not been studied previously.

Binarity: The Next Big Leap in Deepening Our Understanding of (Pulsational) Pair-Instability Supernovae

This work will provide a significant breakthrough in understanding the progenitors of the elusive (P)PISN, with huge ramifications for unraveling the nature of the associated astrophysical transients, their signatures in gravitational wave detections, and ultimately the evolution of very massive stars. To advance our understanding in interpreting observations, we need models that include the effects of binarity to be realistic. Our work will finally tackle the bottleneck question: how do binary interactions affect the pair-instability phenomenon?

This proposed research is both timely and critical to fully, and accurately utilize the potential science expected from rapidly growing observational datasets. **The models that we will develop are directly relevant to advance the SMD Division "Astrophysics Research Program" to "explore the origin and evolution of the galaxies, stars and planets that make up our universe."** Our fully public models will serve multifold interests for the community. This includes interpreting and characterizing the electromagnetic transients associated to (P)PISN, highlighted in Panel B of the 2020 Astrophysical Decadal Survey (National Academies of Sciences and Medicine, 2021) as one of the last theoretically expected astrophysical transients still lacking an uncontroversial observational detection. The panel also highlights that a crucial goal for GW surveys is to probe the black hole mass gap expected from pair-instability supernovae, which has the potential to reveal key insights into nuclear and stellar physics. Thus, the products of this research program will significantly enhance the legacy value of NASA and enable wide-ranging impactful science.

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